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AMIT SINGH PARIHAR

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Github Link; Personal Portfolio; LinkedIn

SUMMARY

GenAI Engineer with 1.5+ years of enterprise experience at TCS, designing and deploying production-grade LLM-powered modernization systems using RAG, agentic workflows, and semantic retrieval pipelines. Specialized in large-scale legacy transformation and automated enterprise documentation using Open-Source LLMs and vector databases.

TECHNICALSKILLS

- **Programming and DB Languages** - Python, MySql, Vector Databases.
- **API –** Flask, FastAPI
- **LLM Tools and Frameworks** - LangChain, LangGraph, RAG, Cross-Encoder Re-ranking, Prompt Engineering, OpenAI APIs, Qwen, Open-source LLMs.
- **DS/ML/AI** - Python, Generative AI, Statistics, Machine Learning, Deep Learning, NLP, DS lifecycle
- **ML Framework/ Visualization** - Sklearn, Numpy, Tensorflow, Pandas, Matplotlib, Seaborn etc.
- **Cloud** – AWS(AWS Bedrock, SageMaker, S3, EC2, Lambda, API Gateway, IAM)

EDUCATION

LNCT University(Bhopal) **Sept 2021 – June 2024**
Bachelor of Computer Applications (BCA)
Specialization in AI/Data Analytics, CGPA: 8.34

EXPERIENCE

ART - Abstinent Research & Technologies – Python Developer Intern **Jan 2024 – July 2024**

Developed an End-to-End LLM Application for Chatting with MySQL Using an LLM -

- Built a natural language-to-SQL system enabling users to query a MySQL database using plain English.
- Applied few-shot prompting techniques to guide the LLM in generating accurate and structured SQL queries.
- Generated and executed SQL queries supporting multi-table joins and filters, reducing incorrect query generation and delivering real-time results.

Tata Consultancy Services (TCS)
GenAI Developer **August 2024 – Present**
Enterprise Modernization & Code Transformation

- Worked on GenAI-based applications to modernize and upgrade large enterprise applications using Large Language Models.
- Developed code conversion solutions for legacy and modern technologies, including Java to Kotlin, VB6 to Python, and framework migrations such as ASP.NET to ASP.NET Core.
- Built GenAI solutions for technology upgrades, including Java version upgrades and React framework modernization.

Java-to-Kotlin Migration Engine

- Led the development of a Java-to-Kotlin conversion tool, covering project scanning, code conversion, dependency handling, and validation to ensure production-ready output, reducing manual migration effort by 60%.
- Used language-aware code chunking and well-defined prompts with Qwen3-Coder to improve conversion accuracy and maintain proper code structure.
- Implemented validation and retry logic where generated code was reviewed and refined to ensure correctness and successful compilation.

Automated Documentation & RAG Systems

- Developed an automated FSD (Functional Specification Document) generation system where application code was converted into embeddings using Qwen3 Embeddings, stored in pgvector, and retrieved with re-ranking, reducing documentation preparation time by 65% to generate accurate, client-ready documentation.
- Built an end-to-end GenAI system to generate BRDs from legacy VB6 code using logic-aware chunking and RAG architecture.
- Created embeddings stored in a vector database with semantic retrieval and cross-encoder re-ranking, improving retrieval precision by 25% and reducing hallucinations.

- Designed context-aware prompting and automated documentation workflows for enterprise modernization.

LICENSES & CERTIFICATIONS

- Data Analytics and Visualization Virtual Experience – Accenture
- British Airways Data Science - British Airways
- AWS Certified Cloud Practitioner – Udemy
- Generative AI, from GANs to CLIP with Python– Udemy
- Python Developer Internship – ART Abstinent Research & Technologies
- Certificate of Accomplishment - Samatrix Consulting Private Limited (Foundation to AI Data Science& Data Analysis)